

MapReduce Algorithm – Part 3

Weather Station Data



Weather Station Data Set

Weather data collected in 1800.

Identify the lowest minimum temperature recorded.

	Weather Station ID ↑	Date ↑	Data Type ↑	Value (if temperature, expressed in tenths of a Celsius degree)	Extra Columns ↑	
	A	B	C	D	E	F
1	ITE00100554	18000101	TMAX	-75		E
2	ITE00100554	18000101	TMIN	-148		E
3	GM000010962	18000101	PRCP	0		E
4	EZE00100082	18000101	TMAX	-86		E
5	EZE00100082	18000101	TMIN	-135		E
6	ITE00100554	18000102	TMAX	-60	I	E
7	ITE00100554	18000102	TMIN	-125		E
8	EZE00100082	18000102	TMIN	-130		E

Historical Climatology. (n.d.) Climate history databases: Starting points for students, teachers, and professors. Retrieved from
URL: <https://www.historicalclimatology.com/databases.html>



MapReduce - Mapper

	A	B	C	D	E	F
1	ITE00100554	18000101	TMAX	-75		E
2	ITE00100554	18000101	TMIN	-148		E
3	GM000010962	18000101	PRCP	0		E
4	EZE00100082	18000101	TMAX	-86		E
5	EZE00100082	18000101	TMIN	-135		E
6	ITE00100554	18000102	TMAX	-60	I	E
7	ITE00100554	18000102	TMIN	-125		E
8	EZE00100082	18000102	TMIN	-130		E

MAPPER

ITE00100554 , 5.36

EZE00100082 , 7.7

ITE00100554 , 9.5

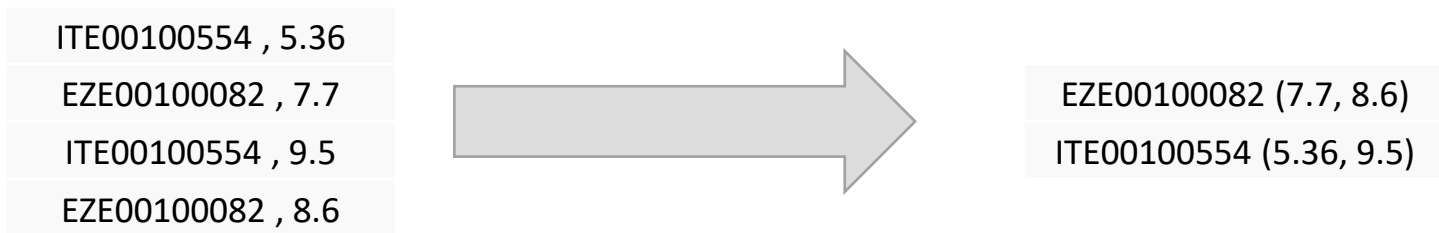
EZE00100082 , 8.6

What is happening within the Mapper?

- Extract and organize the data into Key-Value pairs.



MapReduce – Sort and Shuffle



What is happening in the Sort and Shuffle Stage?

- Take the Key-Value pairs from the Mapper, sorts by keys and groups together all values for that key.



MapReduce – Reducer



What is happening within the Reducer?

- For each key, the Reducer returns the smallest temperature in the list.

